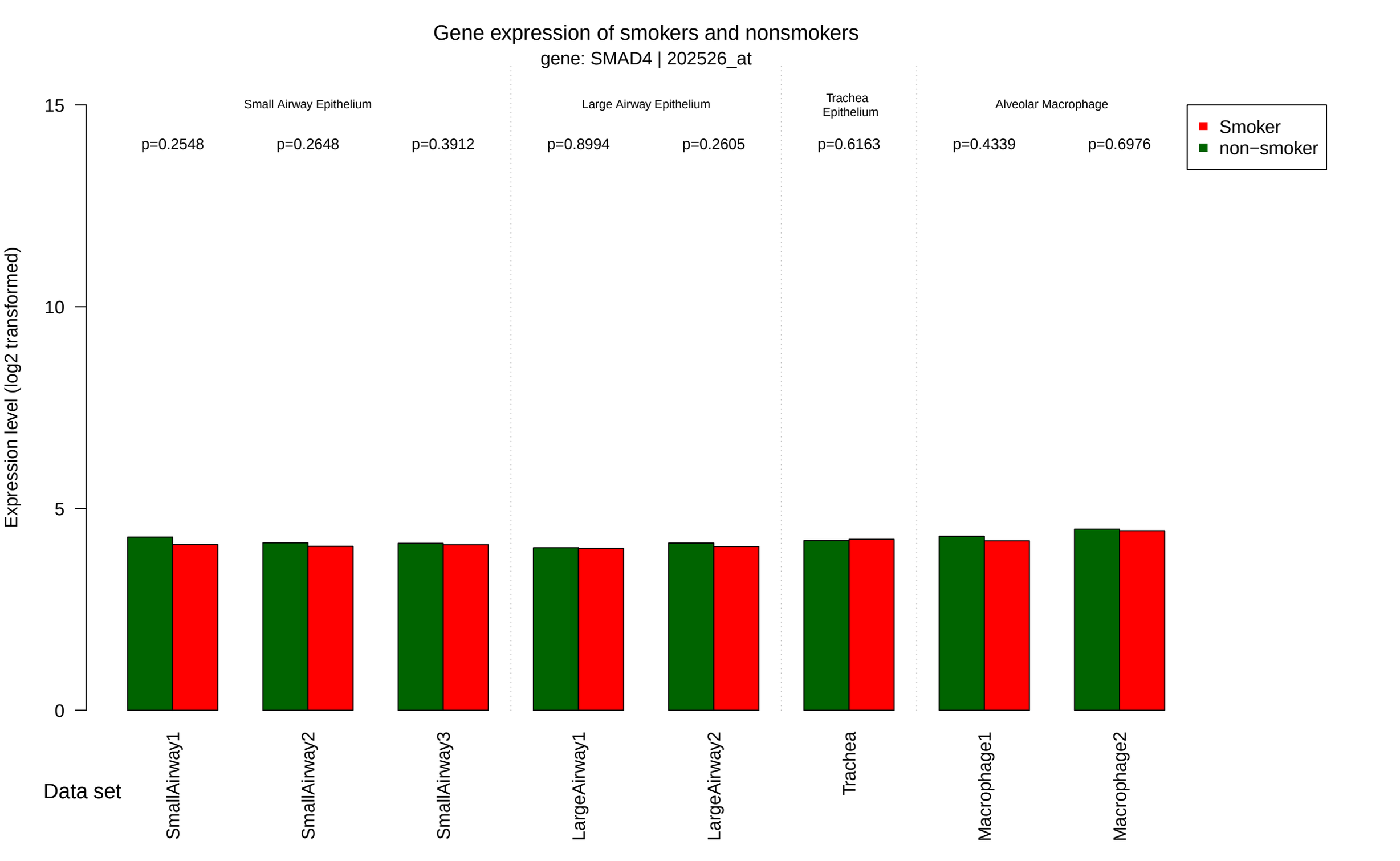
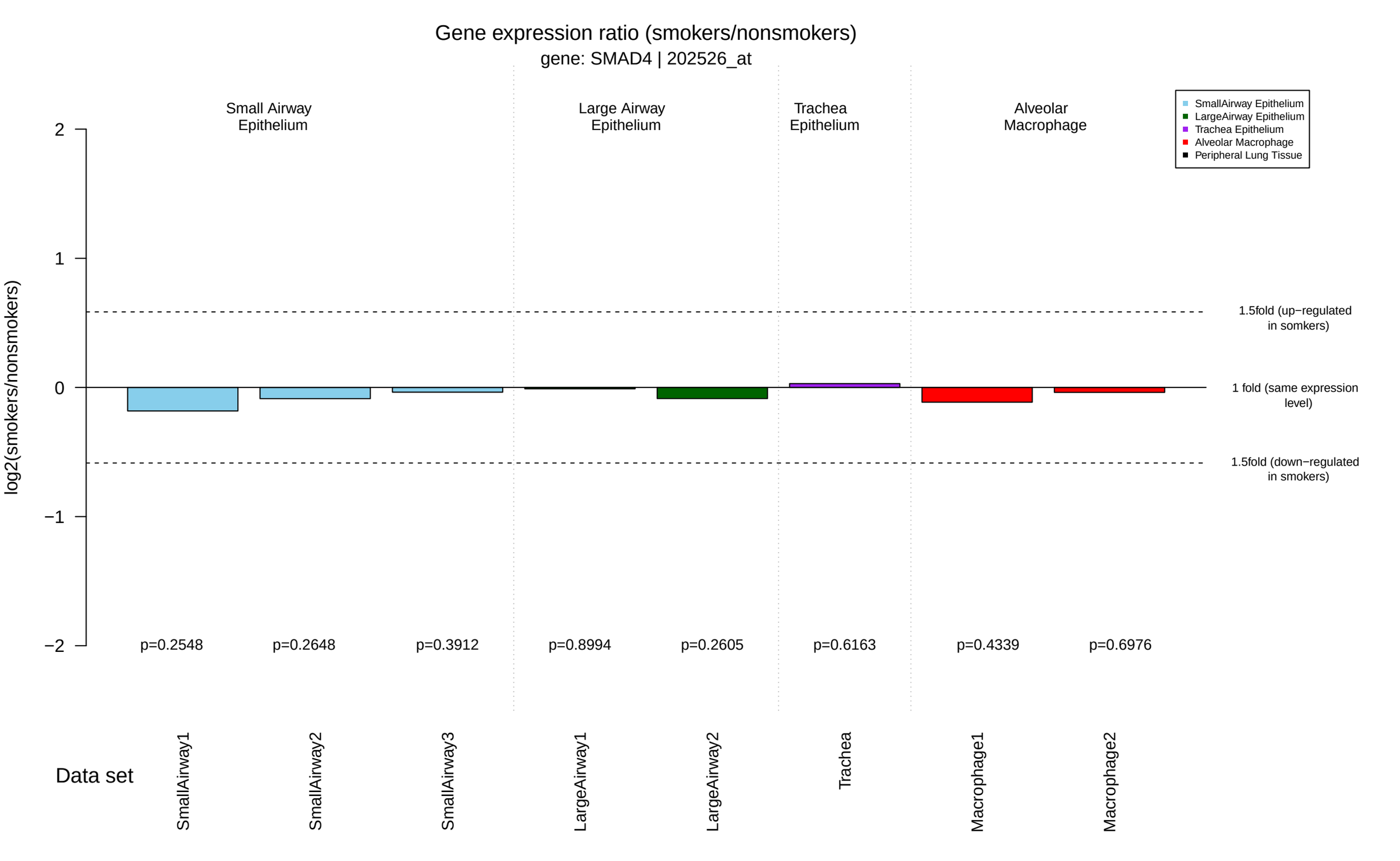
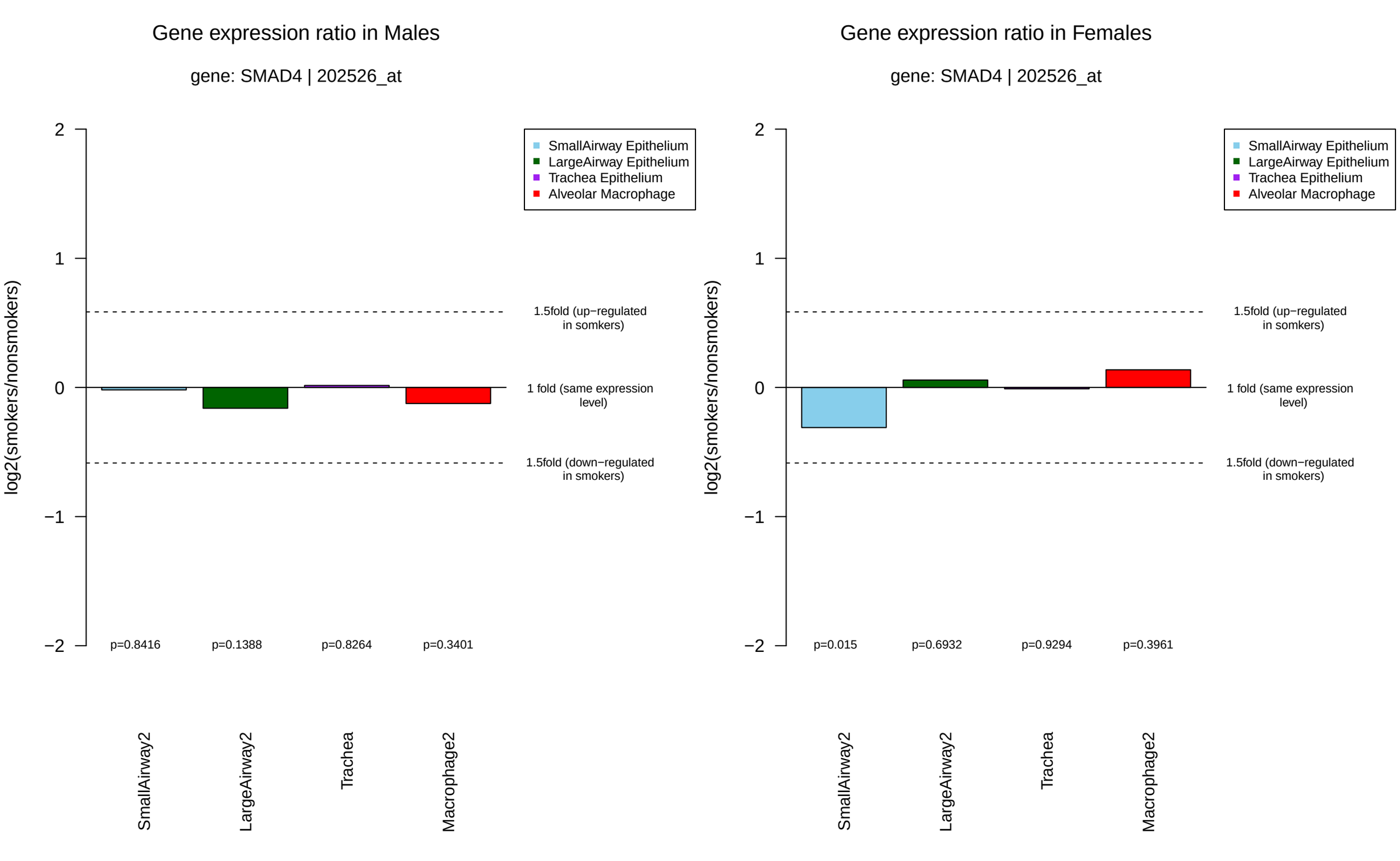
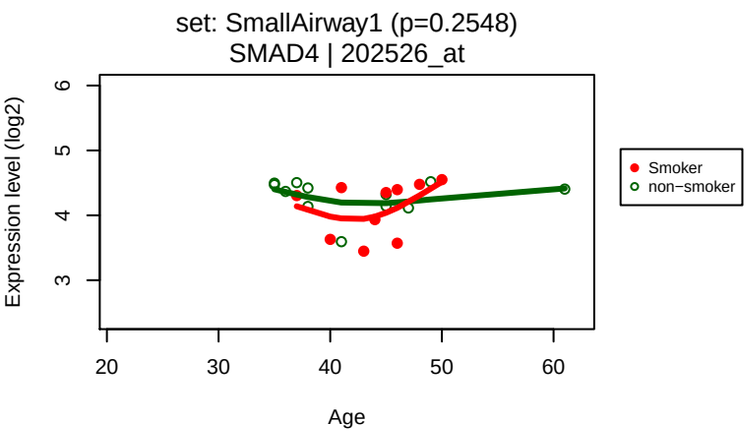
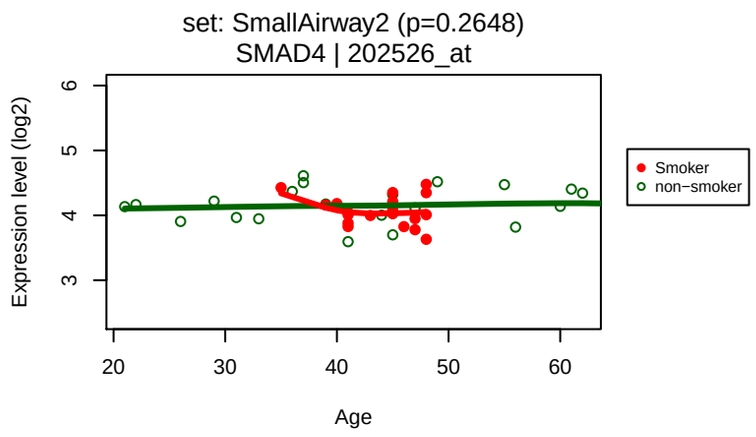
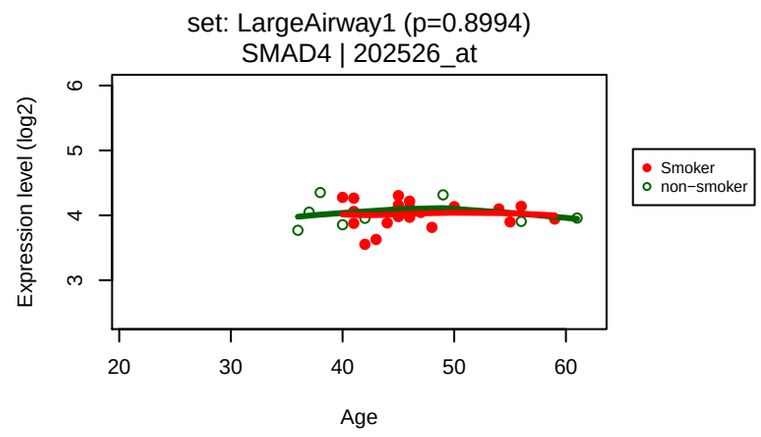

Gene expression of smokers and nonsmokers
gene: SMAD4 | 202526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2548
p=0.2648
p=0.3912
p=0.8994
p=0.2605
p=0.6163
p=0.4339
p=0.6976
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

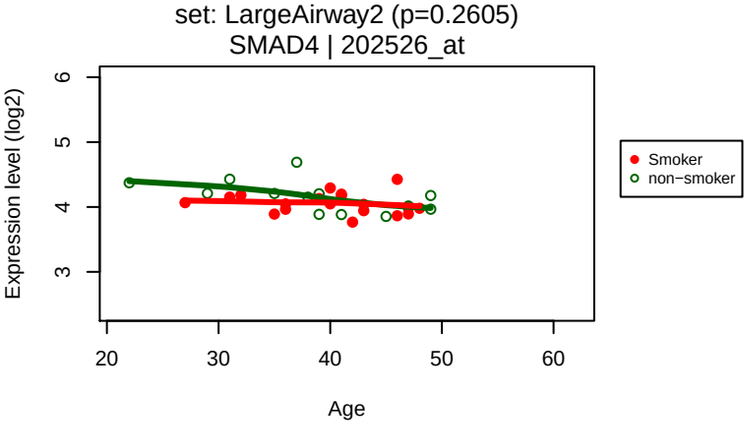

Gene expression ratio (smokers/nonsmokers)
gene: SMAD4 | 202526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2548
p=0.2648
p=0.3912
p=0.8994
p=0.2605
p=0.6163
p=0.4339
p=0.6976
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

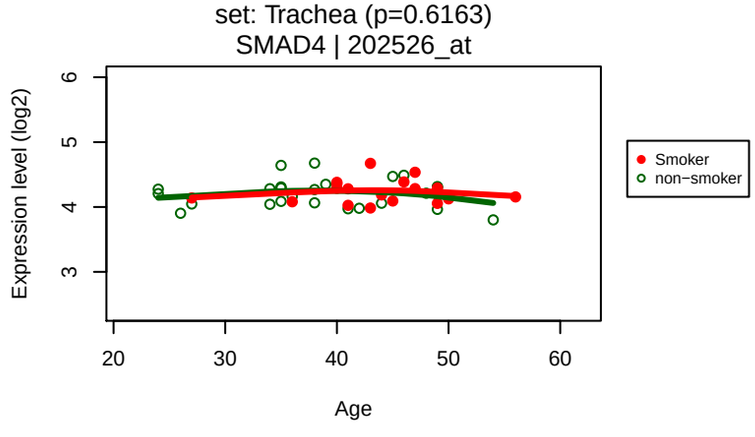

Gene expression ratio in Males
gene: SMAD4 | 202526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8416
p=0.1388
p=0.8264
p=0.3401
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD4 | 202526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.015
p=0.6932
p=0.9294
p=0.3961
SmallAirway2
LargeAirway2
Trachea
Macrophage2

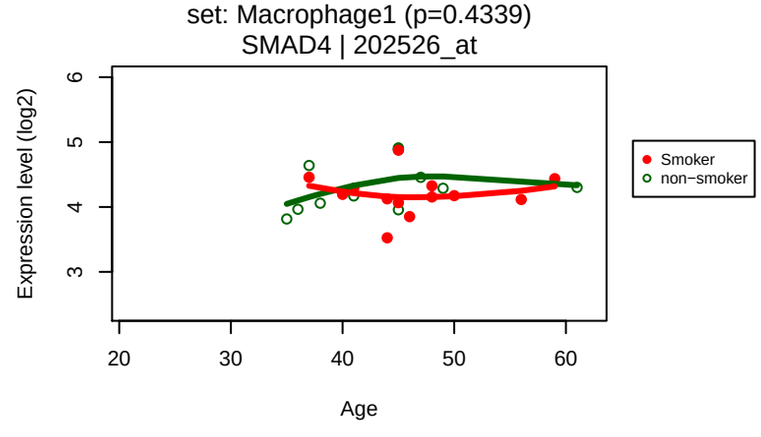

set: SmallAirway1 (p=0.2548)
SMAD4 | 202526_at
Expression level (log2)
Age
Smoker
non−smoker

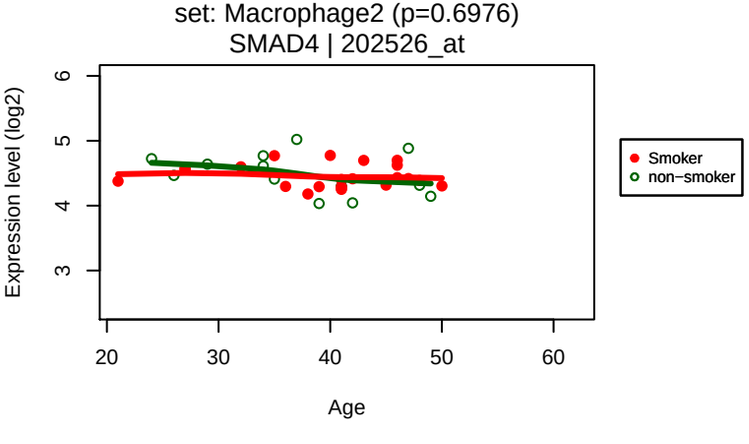

set: SmallAirway2 (p=0.2648)
SMAD4 | 202526_at
Expression level (log2)
Age
Smoker
non−smoker

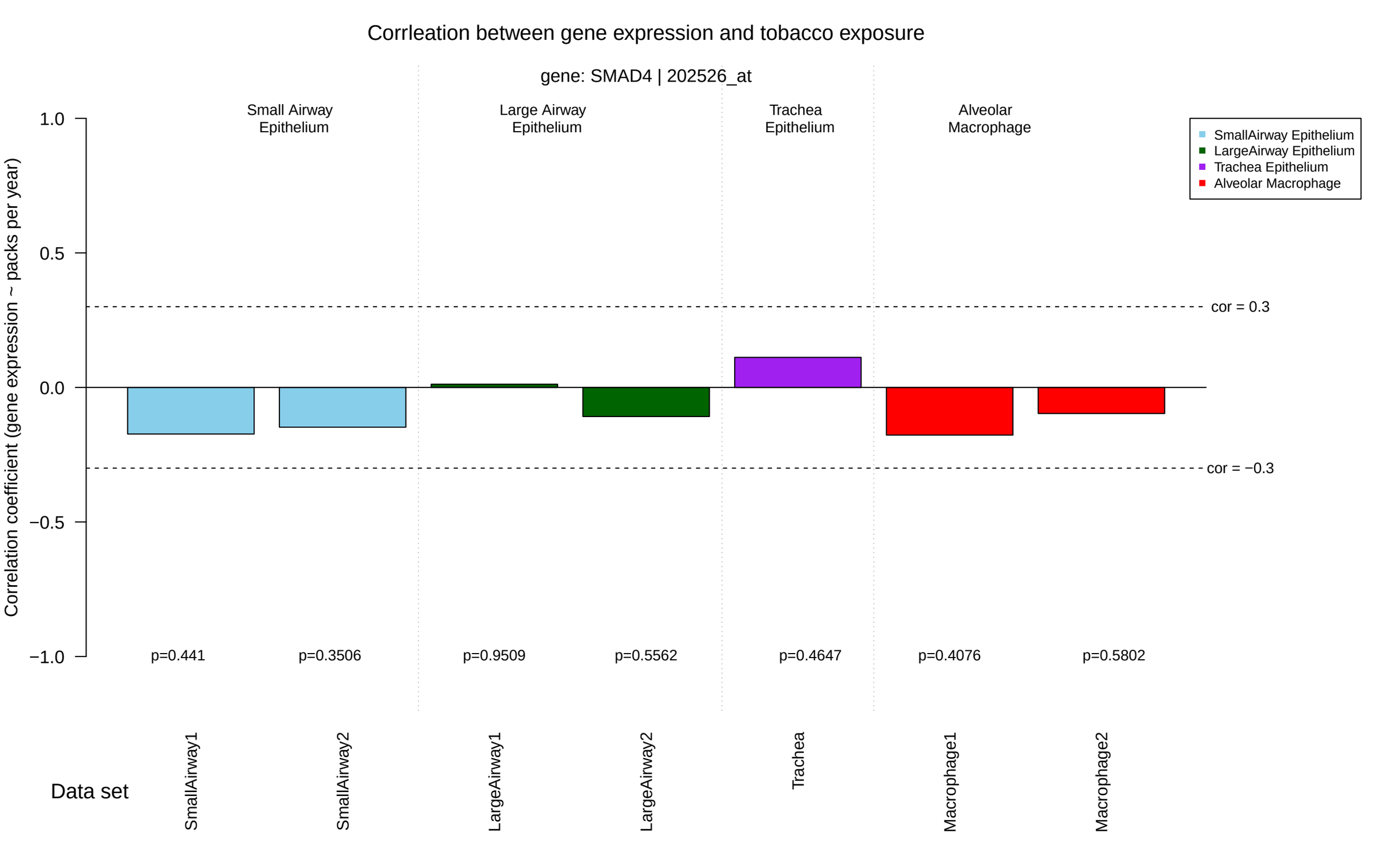

set: LargeAirway1 (p=0.8994)
SMAD4 | 202526_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2605)
SMAD4 | 202526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6163)
SMAD4 | 202526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4339)
SMAD4 | 202526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6976)
SMAD4 | 202526_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD4 | 202526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.441
p=0.3506
p=0.9509
p=0.5562
p=0.4647
p=0.4076
p=0.5802
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2